

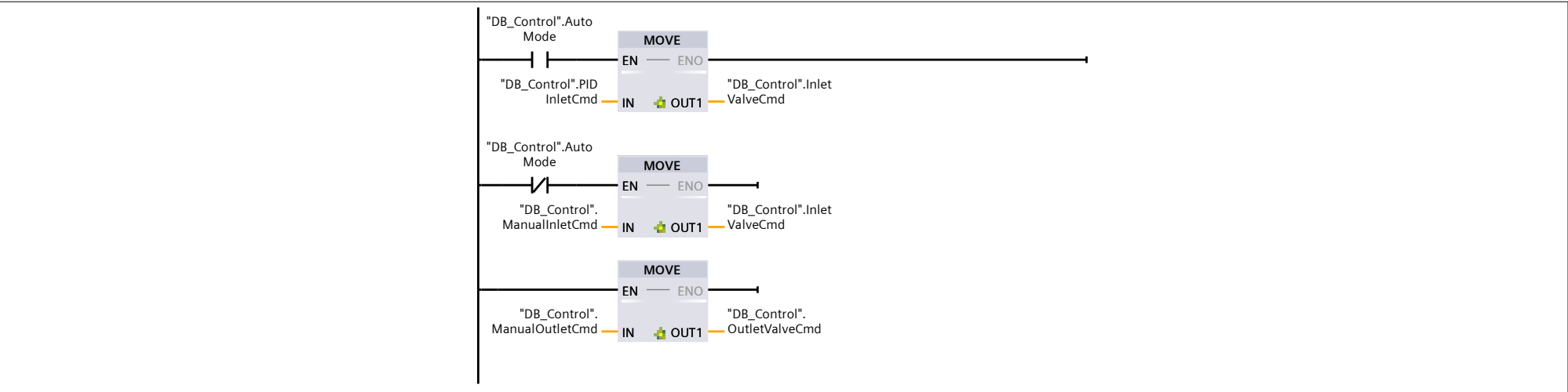
Program blocks

Main [OB1]

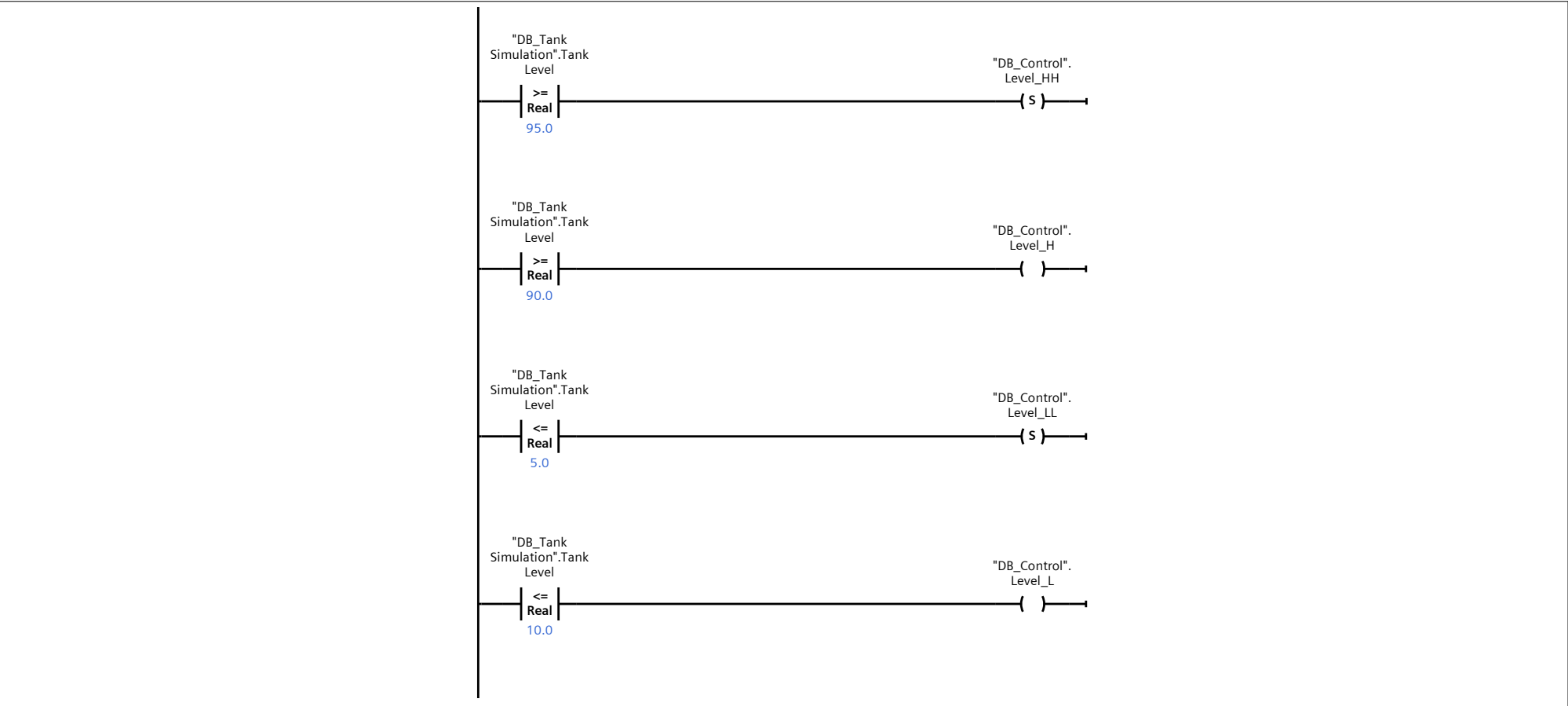
Main Properties							
General							
Name	Main	Number	1	Type	OB	Language	LAD
Numbering	Automatic						
Information							
Title	"Main Program Sweep (Cycle)"	Author		Comment		Family	
Version	0.1	User-defined ID					

Name	Data type	Default value	Comment
▼ Input			
Initial_Call	Bool		Initial call of this OB
Remanence	Bool		=True, if remanent data are available
Temp			
Constant			

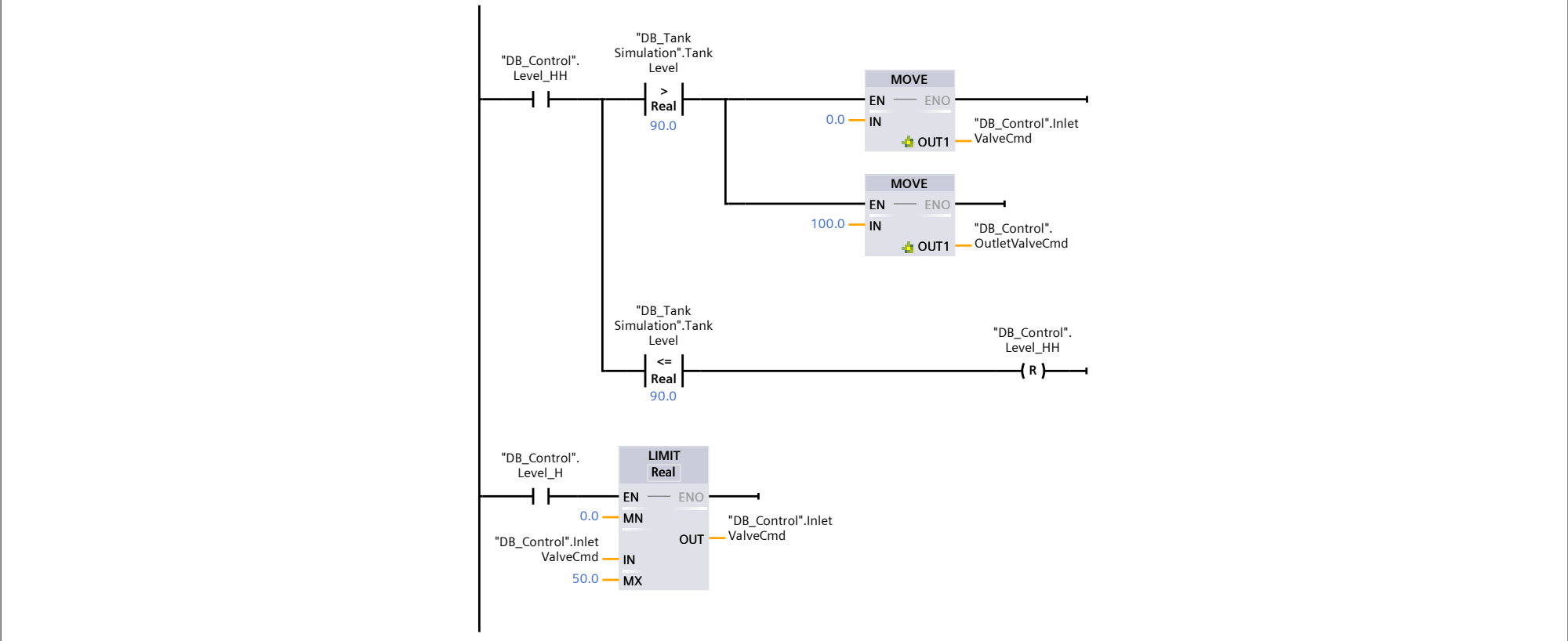
Network 1: Operating mode command selection



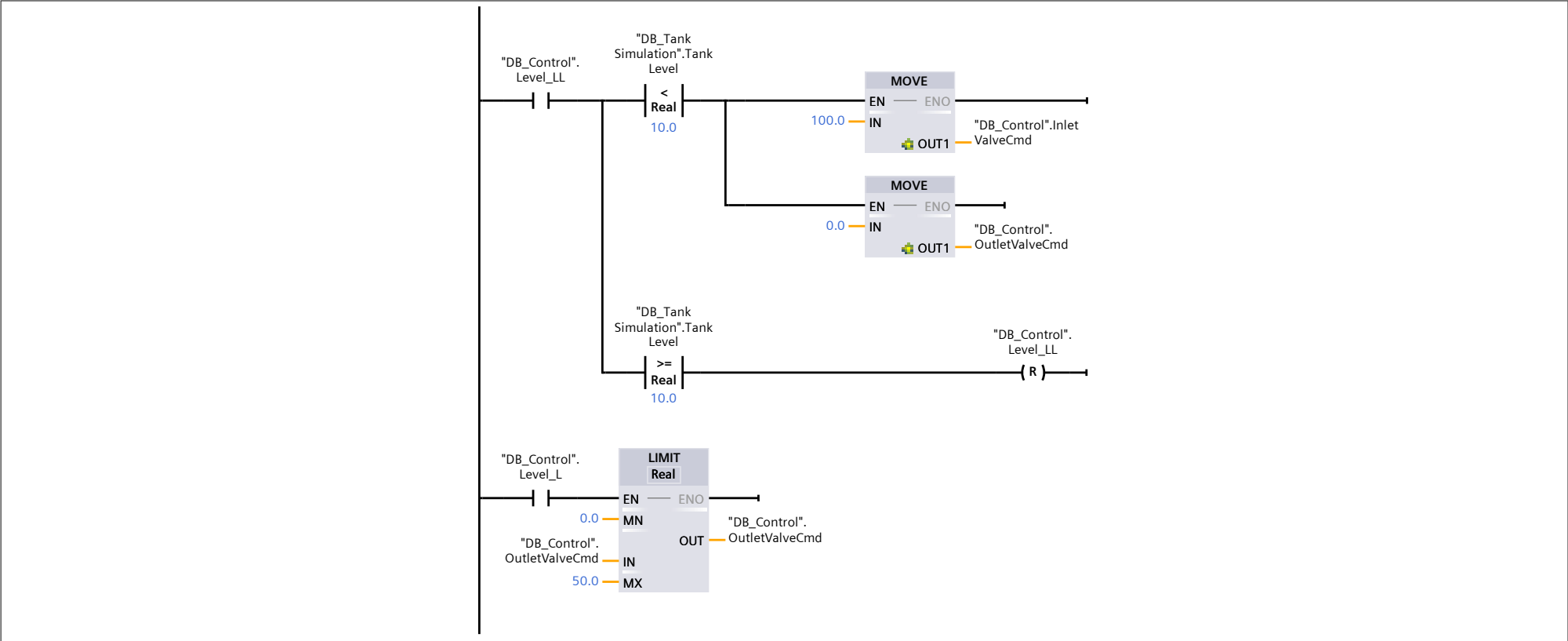
Network 2: Level alarm thresholds



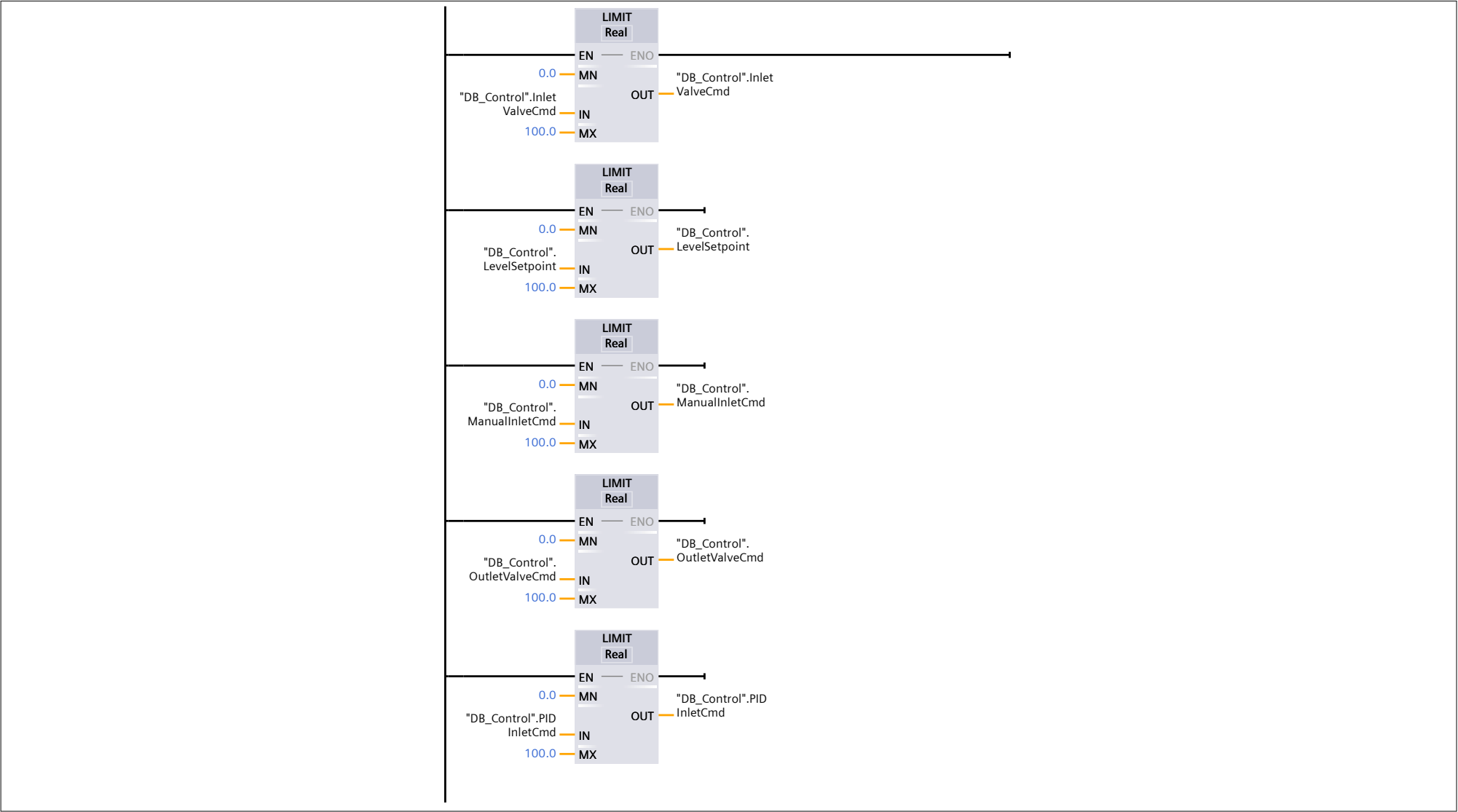
Network 3: High level protection



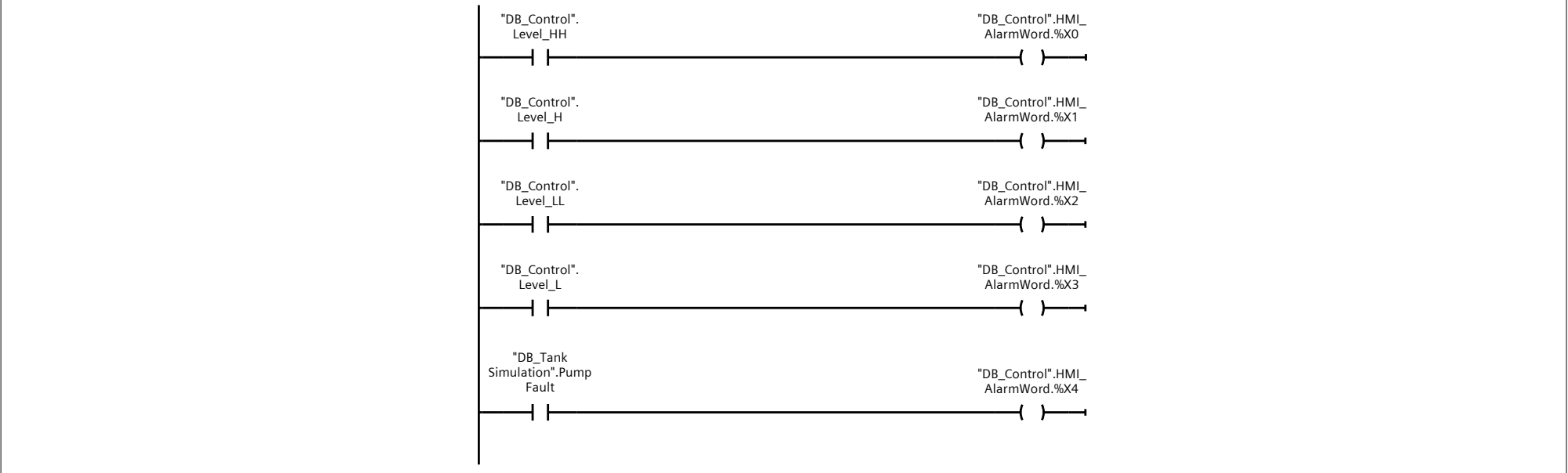
Network 4: Low level protection



Network 5: Command value limits



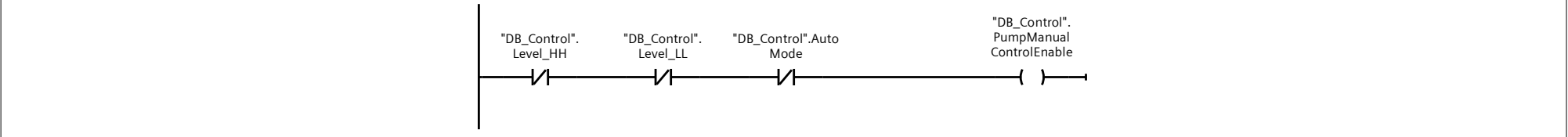
Network 6: HMI alarm word mapping



Network 7: Pump command control

```
0001 IF "DB_Control".Level_HH THEN
0002     "DB_TankSimulation".PumpCmd := FALSE;
0003 ELSIF "DB_Control".Level_LL THEN
0004     "DB_TankSimulation".PumpCmd := TRUE;
0005 ELSIF "DB_Control".AutoMode THEN
0006     "DB_TankSimulation".PumpCmd := TRUE;
0007 ELSE
0008     "DB_TankSimulation".PumpCmd := "DB_Control".ManualPumpCmd;
0009
0010 END_IF;
```

Network 8: Manual pump control enable



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Program blocks

FB_TankSimulation [FB1]

FB_TankSimulation Properties

General

Name	FB_TankSimulation	Number	1	Type	FB	Language	SCL
Numbering	Automatic						

Information

Title		Author		Comment		Family	
Version	0.1	User-defined ID					

Name	Data type	Default value	Retain	Accessible from HMI/OPC UA/Web API	Writ-able from HMI/OPC UA/ Web API	Visible in HMI engi-neering	Setpoint	Supervi-sion	Comment
▼ Input									
CycleTime	Real	0.1	Non-retain	True	True	True	False		
InletValveCmd	Real	0.0	Non-retain	True	True	True	False		
OutletValveCmd	Real	0.0	Non-retain	True	True	True	False		
Output									
InOut									
▼ Static									
TankLevel	Real	0.0	Non-retain	True	True	True	False		
Inflow	Real	0.0	Non-retain	True	True	True	False		
Outflow	Real	0.0	Non-retain	True	True	True	False		
InletValveActual	Real	0.0	Non-retain	True	True	True	False		
OutletValveActual	Real	0.0	Non-retain	True	True	True	False		
ValveSpeed	Real	5.0	Non-retain	True	True	True	False		
PumpRunning	Bool	false	Non-retain	True	True	True	False		
PumpSpeed	Real	2.0	Non-retain	True	True	True	False		
PumpTimer	Real	0.0	Non-retain	True	True	True	False		
PumpFailureSimulation	Bool	false	Non-retain	True	True	True	False		
PumpFault	Bool	false	Non-retain	True	True	True	False		
PumpFaultReset	Bool	false	Non-retain	True	True	True	False		
PumpCmd	Bool	false	Non-retain	True	True	True	False		
Temp									
Constant									

```
0001 REGION Pump_simulation
0002
0003     IF #PumpCmd AND NOT #PumpFault THEN
0004         #PumpTimer := #PumpTimer + (#PumpSpeed * #CycleTime);
0005
0006         IF #PumpTimer >= 2.0 AND NOT #PumpFailureSimulation THEN
0007             #PumpRunning := TRUE;
0008             #PumpTimer := 2.0;
0009
0010         ELSIF #PumpTimer >= 4.0 AND #PumpFailureSimulation THEN
0011             #PumpFault := TRUE;
0012             #PumpTimer := 0.0;
0013         END_IF;
0014
0015     END_IF;
0016
0017     IF NOT #PumpCmd THEN
0018         #PumpTimer := #PumpTimer - (#PumpSpeed * #CycleTime);
0019
0020         IF #PumpTimer <= 0.0 THEN
0021             #PumpRunning := FALSE;
0022             #PumpTimer := 0.0;
0023         END_IF;
0024
0025     END_IF;
0026
0027     IF #PumpFailureSimulation AND #PumpRunning THEN
0028         #PumpRunning := FALSE;
0029         #PumpFault := TRUE;
0030         #PumpTimer := 0.0;
0031     END_IF;
0032
0033     IF #PumpFaultReset AND NOT #PumpFailureSimulation THEN
0034         #PumpFault := FALSE;
0035     END_IF;
0036
0037 END_REGION
0038
0039
0040 REGION Inlet_valve_simulation
0041
0042     IF #InletValveActual < #InletValveCmd THEN
```

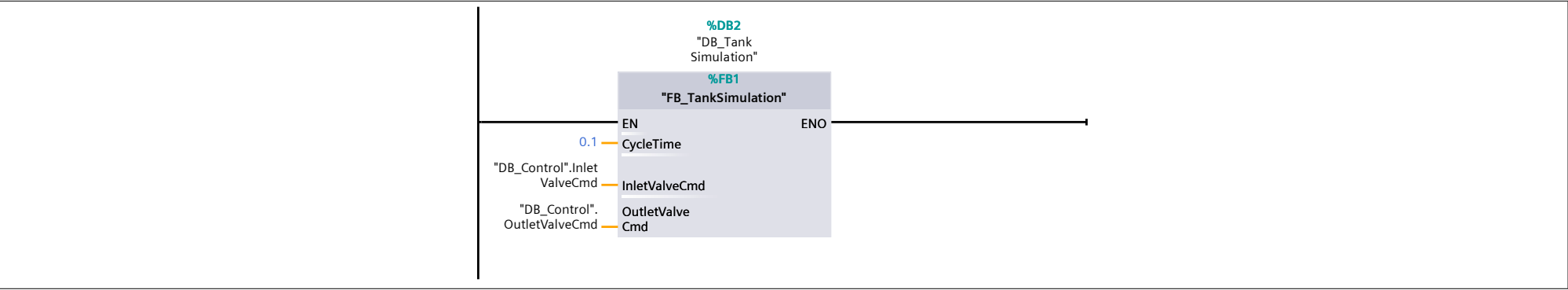
Totally Integrated Automation Portal			
<pre>0043 #InletValveActual := #InletValveActual + (#ValveSpeed * #CycleTime); 0044 0045 IF #InletValveActual > #InletValveCmd THEN 0046 #InletValveActual := #InletValveCmd; 0047 END_IF; 0048 0049 ELSIF #InletValveActual > #InletValveCmd THEN 0050 #InletValveActual := #InletValveActual - (#ValveSpeed * #CycleTime); 0051 0052 IF #InletValveActual < #InletValveCmd THEN 0053 #InletValveActual := #InletValveCmd; 0054 END_IF; 0055 0056 END_IF; 0057 0058 END_REGION 0059 0060 0061 REGION Outlet_valve_simulation 0062 0063 IF #OutletValveActual < #OutletValveCmd THEN 0064 #OutletValveActual := #OutletValveActual + (#ValveSpeed * #CycleTime); 0065 0066 IF #OutletValveActual > #OutletValveCmd THEN 0067 #OutletValveActual := #OutletValveCmd; 0068 END_IF; 0069 0070 ELSIF #OutletValveActual > #OutletValveCmd THEN 0071 #OutletValveActual := #OutletValveActual - (#ValveSpeed * #CycleTime); 0072 0073 IF #OutletValveActual < #OutletValveCmd THEN 0074 #OutletValveActual := #OutletValveCmd; 0075 END_IF; 0076 0077 END_IF; 0078 0079 END_REGION 0080 0081 0082 REGION Flow_calculation 0083 0084 IF #PumpRunning THEN 0085 #Inflow := (#InletValveActual / 100.0) * 2.0; 0086 ELSE 0087 #Inflow := 0.0; 0088 END_IF; 0089 0090 #Outflow := (#OutletValveActual / 100.0) * 2.0; 0091 0092 END_REGION 0093 0094 0095 REGION Tank_level_simulation 0096 0097 #TankLevel := #TankLevel + (#Inflow - #Outflow) * #CycleTime; 0098 0099 IF #TankLevel > 100.0 THEN 0100 #TankLevel := 100.0; 0101 END_IF; 0102 0103 IF #TankLevel < 0.0 THEN 0104 #TankLevel := 0.0; 0105 END_IF; 0106 0107 END_REGION</pre>			
Symbol	Address	Type	Comment
#CycleTime		Real	
#Inflow		Real	
#InletValveActual		Real	
#InletValveCmd		Real	
#Outflow		Real	
#OutletValveActual		Real	
#OutletValveCmd		Real	
#PumpCmd		Bool	
#PumpFailureSimulation		Bool	
#PumpFault		Bool	
#PumpFaultReset		Bool	
#PumpRunning		Bool	
#PumpSpeed		Real	
#PumpTimer		Real	
#TankLevel		Real	
#ValveSpeed		Real	

Program blocks

OB_TankSimulation [OB30]

OB_TankSimulation Properties							
General							
Name	OB_TankSimulation	Number	30	Type	OB	Language	LAD
Numbering	Automatic						
Information							
Title		Author		Comment		Family	
Version	0.1	User-defined ID					
Name		Data type	Default value		Comment		
▼ Input							
Initial_Call		Bool			Initial call of this OB		
Event_Count		Int			Events discarded		
Temp							
Constant							

Network 1: Tank Simulation Execution



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Program blocks

DB_TankSimulation [DB2]

DB_TankSimulation Properties

General

Name	DB_TankSimulation	Number	2	Type	DB	Language	DB
Numbering	Automatic						

Information

Title		Author		Comment		Family	
Version	0.1	User-defined ID					

Name	Data type	Start value	Retain	Accessible from HMI/OPC UA/Web API	Writ-able from HMI/ OPC UA/ Web API	Visible in HMI engi-neering	Setpoint	Supervi-sion	Comment
▼ Input									
CycleTime	Real	0.1	False	True	True	True	False		Simulation cycle time [s]
InletValveCmd	Real	0.0	False	True	True	True	False		Inlet valve position command [%]
OutletValveCmd	Real	0.0	False	True	True	True	False		Outlet valve position command [%]
Output									
InOut									
▼ Static									
TankLevel	Real	0.0	False	True	True	True	False		Simulated tank level [%]
Inflow	Real	0.0	False	True	True	True	False		Calculated inlet flow
Outflow	Real	0.0	False	True	True	True	False		Calculated outlet flow
InletValveActual	Real	0.0	False	True	True	True	False		Simulated inlet valve position [%]
OutletValveActual	Real	0.0	False	True	True	True	False		Simulated outlet valve position [%]
ValveSpeed	Real	5.0	False	True	True	True	False		Valve movement speed [%/s]
PumpRunning	Bool	false	False	True	True	True	False		Simulated pump running feedback
PumpSpeed	Real	2.0	False	True	True	True	False		Pump start/stop simulation rate
PumpTimer	Real	0.0	False	True	True	True	False		Pump start/stop timer
PumpFailureSimulation	Bool	false	False	True	True	True	False		Pump failure simulation command
PumpFault	Bool	false	False	True	True	True	False		Pump fault status
PumpFaultReset	Bool	false	False	True	True	True	False		Pump fault reset command
PumpCmd	Bool	false	False	True	True	True	False		Pump run command

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Program blocks

DB_Control [DB1]

DB_Control Properties

General

Name	DB_Control	Number	1	Type	DB	Language	DB
Numbering	Automatic						

Information

Title		Author		Comment		Family	
Version	0.1	User-defined ID					

Name	Data type	Start value	Retain	Accessible from HMI/OPC UA/Web API	Writ-able from HMI/OPC UA/ Web API	Visible in HMI engi-neering	Setpoint	Supervi-sion	Comment
▼ Static									
AutoMode	Bool	false	False	True	True	True	False		Automatic operating mode
ManualInletCmd	Real	0.0	False	True	True	True	False		Manual inlet valve command [%]
ManualOutletCmd	Real	0.0	False	True	True	True	False		Manual outlet valve command [%]
PIDInletCmd	Real	0.0	False	True	True	True	False		PID controller output [%]
InletValveCmd	Real	0.0	False	True	True	True	False		Final inlet valve command [%]
OutletValveCmd	Real	0.0	False	True	True	True	False		Final outlet valve command [%]
LevelSetpoint	Real	50.0	False	True	True	True	False		Tank level setpoint [%]
Level_H	Bool	false	False	True	True	True	False		High level alarm
Level_HH	Bool	false	False	True	True	True	False		High-high level protection
Level_L	Bool	false	False	True	True	True	False		Low level alarm
Level_LL	Bool	false	False	True	True	True	False		Low-low level protection
HMI_AlarmWord	Word	16#0	False	True	True	True	False		HMI alarm trigger word
ManualPumpCmd	Bool	false	False	True	True	True	False		Manual pump command
PumpManualControlEnable	Bool	false	False	True	True	True	False		Manual pump control permission

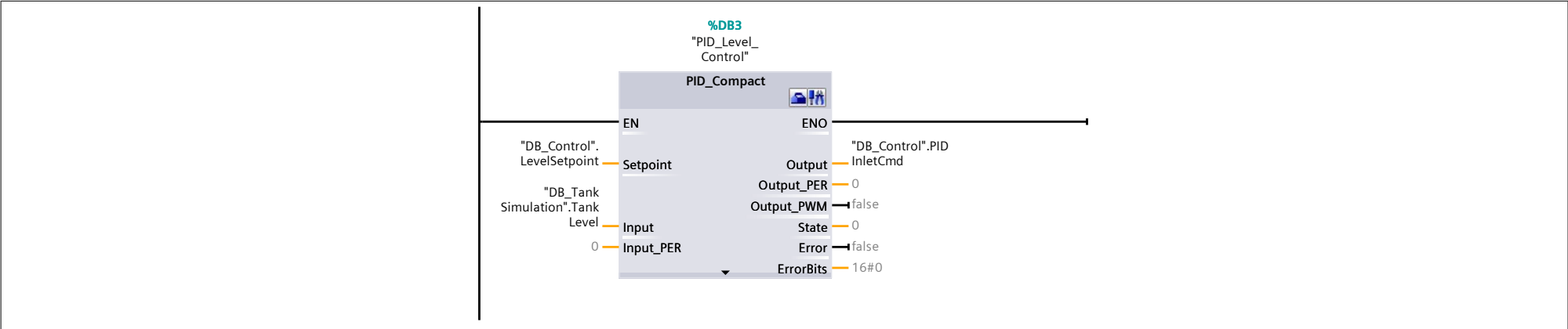
Program blocks

OB_PID_Control [OB31]

OB_PID_Control Properties							
General							
Name	OB_PID_Control	Number	31	Type	OB	Language	LAD
Numbering	Automatic						
Information							
Title		Author		Comment		Family	
Version	0.1	User-defined ID					

Name	Data type	Default value	Comment
▼ Input			
Initial_Call	Bool		Initial call of this OB
Event_Count	Int		Events discarded
Temp			
Constant			

Network 1: PID Controller Execution



PID_Compact [FB1130]

Name	Data type	Default value	Retain	Accessible from HMI/OPC UA/Web API	Writ-able from HMI/OPC UA/ Web API	Visible in HMI engi-neering	Setpoint	Supervi-sion	Comment
▼ Input									
Setpoint	Real	0.0	Non-retain	True	True	True	False		controller setpoint input
Input	Real	0.0	Non-retain	True	True	True	False		current value from process in REAL format
Input_PER	Int	0	Non-retain	True	True	True	False		current value from peripheral input
Disturbance	Real	0.0	Non-retain	True	True	True	False		disturbance intrusion
ManualEnable	Bool	false	Non-retain	True	True	True	False		activate manual value to overwrite output value
ManualValue	Real	0.0	Non-retain	True	True	True	False		manual value
ErrorAck	Bool	false	Non-retain	True	True	True	False		reset error message
Reset	Bool	false	Non-retain	True	True	True	False		reset the controller
ModeActivate	Bool	false	Non-retain	True	True	True	False		enable mode
▼ Output									
ScaledInput	Real	0.0	Non-retain	True	False	True	False		current value after scaling
Output	Real	0.0	Non-retain	True	False	True	False		output value in REAL format
Output_PER	Int	0	Non-retain	True	False	True	False		analog output value
Output_PWM	Bool	false	Non-retain	True	False	True	False		pulse width modulated output value
SetpointLimit_H	Bool	false	Non-retain	True	False	True	False		setpoint reached upper limit
SetpointLimit_L	Bool	false	Non-retain	True	False	True	False		setpoint reached lower limit
InputWarning_H	Bool	false	Non-retain	True	False	True	False		current value reached upper warning level
InputWarning_L	Bool	false	Non-retain	True	False	True	False		current value reached lower warning level
State	Int	0	Non-retain	True	False	True	False		current mode of operation (0-Inactive, 1-SUT, 2-TIR, 3-Automatic, 4-Manual, 5-Substitute output)
Error	Bool	false	Non-retain	True	False	True	False		error flag
ErrorBits	DWord	16#0	Retain	True	False	True	False		error message
▼ InOut									
Mode	Int	4	Retain	True	True	True	False		mode selection
▼ Static									
InternalDiagnostic	DWord	0	Non-retain	False	False	False	False		internal diagnostic and version handling
InternalVersion	DWord	DW#16#02030003	Non-retain	True	False	True	False		version of controller
InternalRTVersion	DWord	0	Non-retain	False	False	False	False		version of runtime
IntegralResetMode	Int	4	Non-retain	True	True	True	True		0 smooth, 1 clear, 2 keep, 3 overwrite initial output, 4 like setpoint change
OverwriteInitialOutputValue	Real	0.0	Non-retain	True	True	True	False		initialisation of output value for override control
RunModeByStartup	Bool	true	Non-retain	True	True	True	True		activate Mode after CPU re-start
LoadBackUp	Bool	false	Non-retain	True	True	True	False		restore last parameter set
SetSubstituteOutput	Bool	true	Non-retain	True	True	True	True		assignment of output value in State = 5 (FALSE - last valid value, TRUE - SubstituteOutput)
PhysicalUnit	Int	0	Non-retain	True	False	True	True		unit of measurement of the process value and setpoint
PhysicalQuantity	Int	0	Non-retain	True	False	True	True		physical quantity of the process value and setpoint
ActivateRecoverMode	Bool	true	Non-retain	True	True	True	True		FALSE - go to inactive by error, TRUE - activate error treatment
Warning	DWord	16#0	Retain	True	False	True	False		warning message
WarningInternal	DWord	16#0	Retain	True	False	True	False		warning message
Progress	Real	0.0	Non-retain	True	False	True	False		progress of current phase in percent
CurrentSetpoint	Real	0.0	Non-retain	True	False	True	False		current active setpoint value
CancelTuningLevel	Real	10.0	Non-retain	True	True	True	True		cancel level for setpoint change during tuning

